

Assignment 1 · Questions

Questions your multi-agent Stock Research Desk must answer

How to use this sheet. Run each question through your pipeline:

```
python main.py --ticker <TICKER> --question "<QUESTION>"
```

For every question, paste into `answers.md`: (1) the agent steps, (2) the final answer, (3) your 1–2 line review. The **"A good answer must..."** column tells you what the grader checks. Prices are live, so there is no single correct number; what matters is that the answer uses the *real* data your agents fetched.

Part A: Market data (tests the Market Data Agent)

#	Ticker	Question	A good answer must...
A1	AAPL	What is Apple's current share price and how much did it move today?	Quote the current price and the % change from the previous close.
A2	MSFT	How close is Microsoft to its 52-week high?	Give the 52-week high and low and say how far (in % or \$) the price is from the high.
A3	TSLA	Which sector and industry is Tesla in, and how big is the company?	Name the sector and industry and give the market cap in billions.
A4	ORCL	Is Oracle expensive based on its P/E ratio?	Quote the P/E. Note that one number alone cannot fully answer 'expensive'.

Part B: Risk (tests the Risk Analyst Agent)

#	Ticker	Question	A good answer must...
B1	NVDA	How risky is NVIDIA compared to a typical stock?	Quote volatility and risk level; explain that above 40% counts as HIGH and 20–40% as MEDIUM.
B2	KO	Is Coca-Cola a low-risk stock for a retiree?	Use volatility and max drawdown; reach a clear yes or no with reasons.
B3	AMZN	What is the worst loss an investor could have suffered in Amazon over the last year?	Use max drawdown and explain it in plain English (peak to trough).
B4	META	Did Meta's 1-year return make up for its risk?	Compare the 1-year return with volatility; give a reasoned opinion.

Part C: Recommendation (tests the Report Writer Agent)

#	Ticker	Question	A good answer must...
C1	GOOGL	Should I buy Alphabet today? Give me a verdict.	End with Verdict: BUY / HOLD / SELL, backed by at least 3 numbers, plus the disclaimer.
C2	JPM	I am a cautious investor with a 1-year horizon. Is JPMorgan suitable for me?	Tie the risk numbers to the investor profile, not just a generic verdict.
C3	INFY	Summarise Infosys for a beginner in 5 bullet points.	Follow the requested format; explain terms like volatility simply.

Part D: Comparison (run the pipeline once per ticker, or do Extension C)

If you have not done Extension C, run both tickers separately and write the comparison yourself from the two outputs.

#	Ticker	Question	A good answer must...
D1	AAPL vs MSFT	Which of the two was less volatile over the last year?	Give both volatility figures and name the less volatile one.
D2	TSLA vs TM	Which had the larger drawdown, Tesla or Toyota?	Give both drawdowns; one sentence on why an EV pure-play might swing more.

Part E: Edge cases (tests robustness and honesty)

These are the most important questions. A good agentic system must **not** make up an answer.

#	Ticker	Question	A good answer must...
E1	XYZ123	What is the share price of XYZ123?	The graph must NOT crash. The answer should say the ticker is invalid or no data was found.
E2	AAPL	What will Apple's share price be next Friday?	Must refuse to predict an exact price; may describe current risk and range instead.
E3	AAPL	Who is the CEO of Apple and what is their salary?	Must say the fetched data does not include this, without guessing.
E4	RELIANCE.N S	What is Reliance Industries' price, and in which currency?	Works for an NSE ticker; notices the price is in INR, not USD.

Part F: Explain your graph (written, no pipeline run)

#	Question
F1	Paste the agent steps for B1. Which node ran the most times, and why?
F2	In E1, which agents still ran even though the ticker was invalid? Suggest a change to the supervisor so the graph stops early.
F3	Why is the supervisor plain Python instead of an LLM? Give one advantage and one limitation.
F4	What would break if steps were declared as a plain list[str] without operator.add? Try it and describe what you see.

answers.md template

```

## A1 - AAPL - What is Apple's current share price ...

**Agent steps**
SUPERVISOR -> market_data_agent
...

**Answer**
<paste the final answer>

**My review**
Correct price cited (checked on Yahoo Finance). No made-up numbers.
```